



Heuristic Methods

BLAST

- It is **Basic local alignment search tool**.
- It is an algorithm for comparing primary biological sequence information, such as the amino-acid sequences of different proteins or the nucleotides of DNA sequences.
- A BLAST search enables a researcher to **compare a query sequence with a library or database of sequences**.

→ *Types of BLAST:*

- Nucleotide-nucleotide BLAST (blastn) :This program, given a DNA query, returns the most similar DNA sequences from the DNA database that the user specifies.
- Protein-protein BLAST (blastp) This program, given a protein query, returns the most similar protein sequences from the protein database that the user specifies.
- Position-Specific Iterative BLAST (PSI-BLAST): This program is used to find **distant relatives of a protein**. PSI-BLAST is much more sensitive in picking up **distant evolutionary relationships** than a standard protein-protein BLAST.

- Nucleotide 6-frame translation-protein (blastx): This program compares the six-frame of a **nucleotide query sequence (both strands)** against a **protein sequence database**. **E.g., if you have a genomic DNA sequence and want to find out which proteins it might encode, you can use blastx.**
- Nucleotide 6-frame translation (tblastx): It translates the query nucleotide sequence in all six possible frames and compares it against the six-frame translations of a nucleotide sequence database. The purpose of tblastx is to **find very distant relationships between nucleotide sequences**. It can help identify homologous sequences that may not share high levels of similarity due to divergence over time.
- Protein-nucleotide 6-frame translation (tblastn): This program compares a protein query against the all six reading frames of a nucleotide sequence database. E.g., tblastn is useful when you have a known protein sequence and want to find potential coding regions in nucleotide sequences (like genomic DNA) that might encode similar proteins. This is helpful in **gene prediction and annotation tasks**.
- Large numbers of query sequences (megablast): When comparing large numbers of input sequences via the command-line BLAST, "megablast" is much faster than running BLAST multiple times. E.g., effective for quickly identifying closely related sequences in large databases

- Go to BLAST tool
- Select type of blast according to query sequence
- Upload file or paste the query sequence.
- Run the blast

Process

- To run the software, BLAST requires a query sequence to search for, and a sequence to search against (also called the target sequence) or a sequence database containing multiple such sequences.
- BLAST will find sub-sequences in the database which are similar to sub sequences in the query. In typical usage, the query sequence is much smaller than the database, e.g., the query may be one thousand nucleotides while the database is several billion nucleotides.
- The main idea of BLAST is that there are often High-scoring Segment Pairs (HSP) contained in a statistically significant alignment.
- BLAST searches for high scoring sequence alignments between the query sequence and the existing sequences in the database using a heuristic approach that approximates the Smith-Waterman algorithm.

BLAST Algorithm

An overview of the BLAST algorithm (a protein to protein search) is as follows:

1) Remove low-complexity region or sequence repeats in the query sequence.

- "Low-complexity region" means a region of a sequence composed of few kinds of elements.
- These regions might give high scores that confuse the program to find the actual significant sequences in the database, so they should be filtered out. The **regions will be marked with an X** (protein sequences) or **N (nucleic acid sequences)** and then be ignored by the BLAST program.
- To filter out the low-complexity regions, the **SEG program** is used for protein sequences and the **program DUST** is used for DNA sequences.

2) Make a k -letter word list of the query sequence.

- Query sequence is split into shorter pieces/words. Take $k=3$ for example, we list the words of length 3 in the query protein sequence "sequentially", until the last letter of the query sequence is included.

3) List the possible matching words.

- This step is one of the main differences between BLAST and FASTA. FASTA cares about all of the common words in the database and query sequences that are listed in step 2; however, BLAST only cares about the high-scoring words.
- The scores are created by comparing the word in the list in step 2 with all the 3-letter words. By using the scoring matrix (substitution matrix) to score the comparison of each residue pair, there are 20^3 possible match scores for a 3-letter word.
- For example, the score obtained by comparing PQG with PQG and PQA is respectively 15 and 7 with the BLOSUM62 weighting scheme. For DNA words, a match is scored as +5 and a mismatch as -3.
- After that, a neighborhood word score threshold T is used to reduce the number of possible matching words. The words whose scores are greater than the threshold T will remain in the possible matching words list, while those with lower scores will be discarded. For example, PQG is kept, but PQA is abandoned when T is 13.
- ***Summary: Now, BLAST checks which of these small 3-letter words (like "PQG") could have good matches in the database sequences. Unlike other programs, BLAST doesn't consider every possible match. It focuses on matches with high scores, which are calculated by comparing the words using a scoring system (like the BLOSUM62 matrix). This scoring matrix assigns points to matches and mismatches based on how often pairs of amino acids are seen together in similar proteins.***

4) Organize the remaining high-scoring words into an efficient search tree.

- This allows the program to rapidly compare the high-scoring words to the database sequences.

5) Repeat step 3 to 4 for each k -letter word in the query sequence.

6) Scan the database sequences for exact matches with the remaining high-scoring words.

- The BLAST program scans the database sequences for the remaining high-scoring word, such as PQG, of each position.

7) Extend the exact matches to high-scoring segment pair (HSP).

- The original version of BLAST stretches a longer alignment between the query and the database sequence in the left and right directions, from the position where the exact match occurred.
- The extension does not stop until the accumulated total score of the HSP begins to decrease.

8) List all of the HSPs in the database whose score is high enough to be considered.

- We list the HSPs whose scores are greater than the empirically determined cutoff score S .

9) Evaluate the significance of the HSP score.

10) Make two or more HSP regions into a longer alignment.

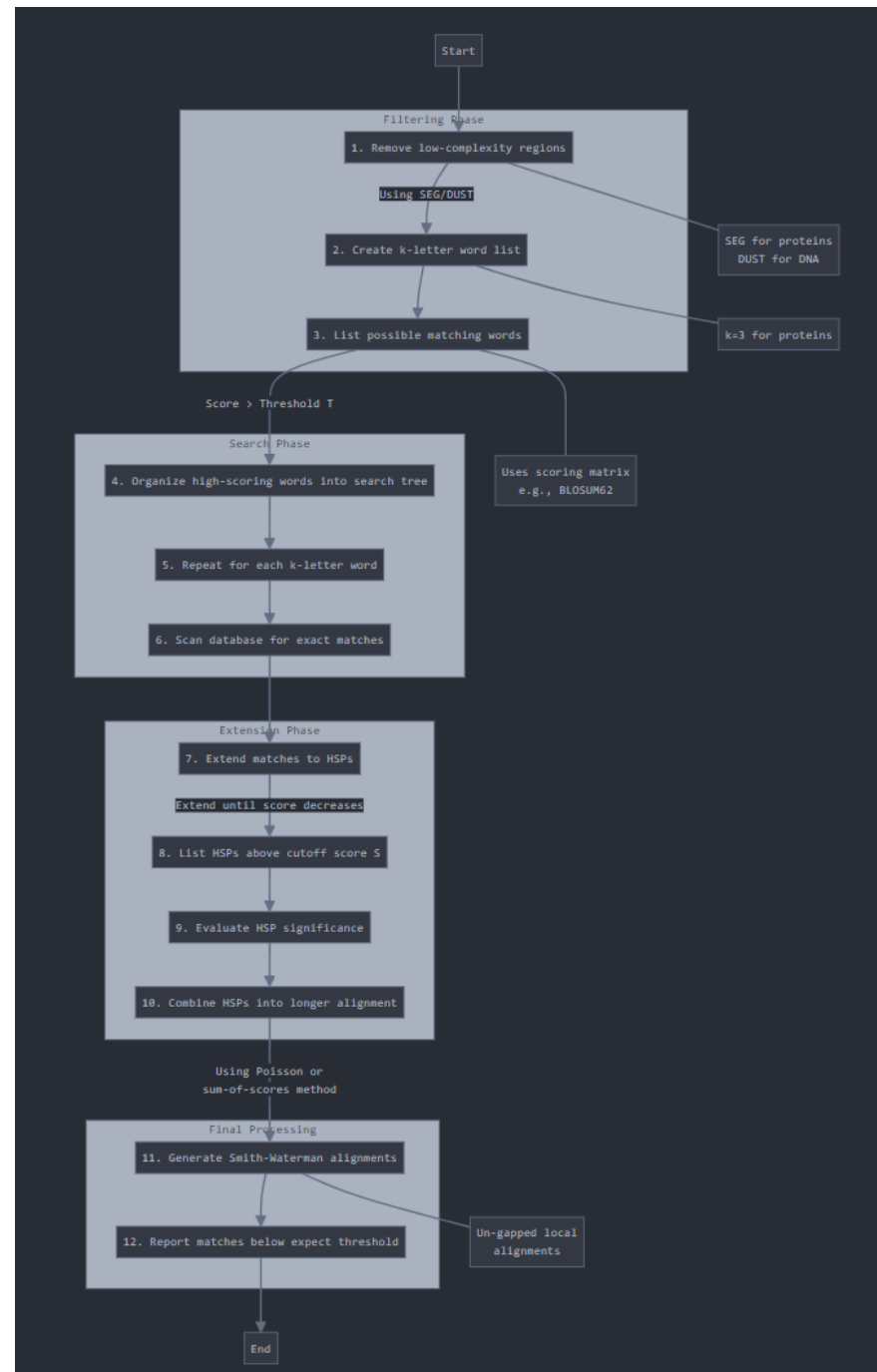
- Sometimes, we find two or more HSP regions in one database sequence that can be made into a longer alignment.
- This provides additional evidence of the relation between the query and database sequence. There are two methods, the Poisson method and the sum-of-scores method, to compare the significance of the newly combined HSP regions.

11) Show the gapped Smith-Waterman local alignments of the query and each of the matched database sequences.

- The original BLAST only generates un-gapped alignments including the initially found HSPs individually, even when there is more than one HSP found in one database sequence.

12) Report every match whose expect score is lower than a threshold parameter.

BLAST Algorithm Summary



FASTA

- FASTA is a pair-wise sequence alignment tool which takes input as nucleotide or protein sequences and compares it with existing databases.
- It is a text-based format and can be read and written with the help of text editor or word processor.
- Fasta file description starts with ‘>’ symbol and followed by the gi and accession number and then the description, all in a single line.

- Next line starts with the sequence and in each row there would be 60 nucleotides/amino acids only.
- For DNA and proteins it is represented in one letter IUPAC nucleotide codes and amino acid codes.
- It finds the local similarity between the sequences and calculates the statistical significance of matches.
- It can be also used to find the functional and evolutionary relationship between the sequences.

FASTA Programs

- FASTA: Compares the protein sequence to another protein sequence in a database or compares nucleotide sequence to another nucleotide sequence in a database.
- FASTX, FASTY: It performs a search for **comparing the nucleotide sequence to a protein sequence database**.
- SSEARCH: It performs a Smith-Watermann alignment, between a protein sequence and another protein sequence/nucleotide sequence and another nucleotide sequence. It is **local alignment**.

- GGSEARCH: Compares a protein or DNA sequence to a sequence database using **Global alignment**. It compares the query sequences that are between 80% of the length of the query.
- GLSEARCH: Compares a protein or DNA sequence to a sequence in a database. The alignments are global in query and local in database. E.g., **You have a complete query sequence (like a protein or gene) and want to see if it matches a specific domain or region within larger, multi-domain sequences in a database.**

FASTA algorithm

- It is derived from logic of the dot plot.
- compute best diagonals from all frames of alignment

The method looks for exact matches between words in query and test sequence

- DNA words are usually 6 nucleotides long
- protein words are 2 amino acids long

FASTA algorithm has 5 steps:

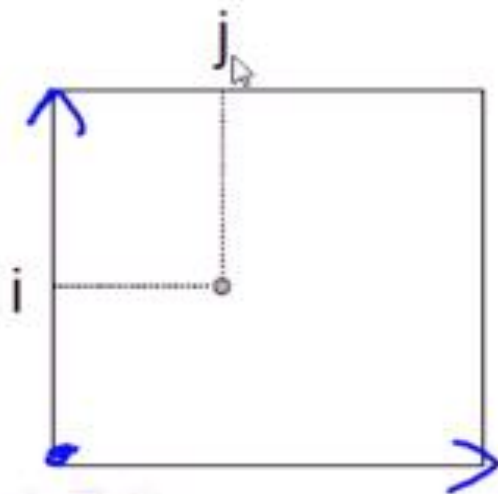
- ***Identify common words between I and J:***
 - Query seq and Database seq Placed on grid.
 - Dot plot
- ***Score diagonals with k words matches, identify 10 best diagonals:***
 - Each diagonal is scored based on the number and quality of k-tuple matches it contains. 10 regions filtered
- ***Rescore initial regions with substitution score matrix.***
 - FASTA refines the scoring of these 10 regions using a substitution matrix (e.g., BLOSUM or PAM for proteins) that assigns scores based on biological similarity between residues.
 - Favors biological meaningful substitutions (high score for biologically meaningful)
- ***Join initial regions using gaps, penalise for gap.***
 - The algorithm then tries to join adjacent high-scoring regions into longer segments by allowing gaps (insertions or deletions) between them.
- ***Perform dynamic programming to find final alignments.***
 - FASTA uses dynamic programming (like the Smith-Waterman algorithm) around the best alignment regions to refine the alignment fully.
 - To get accurate matches, mismatches and placement of gaps.

1. Identifying common k Words between I and J

Word matches in two sequences I and J can be represented as a *dot matrix*

Dot matrix element (i, j) has "a dot", if the word starting at position i in I is identical to the word starting at position j in J

The dot matrix can be plotted for various k

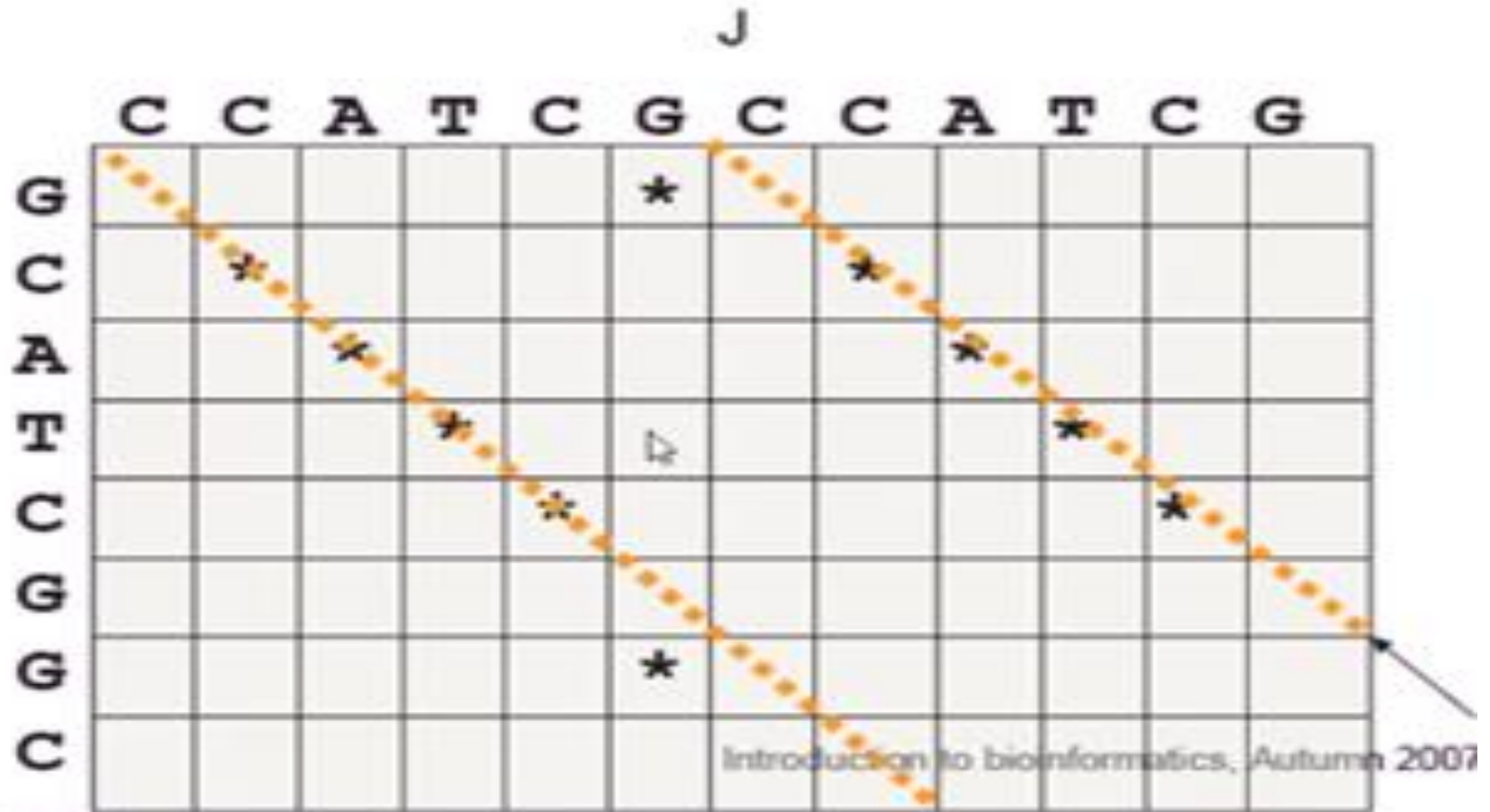


I	=	...	ATCG	i	GATCA	...
J	=	...	TGGT	j	GTCGC	...

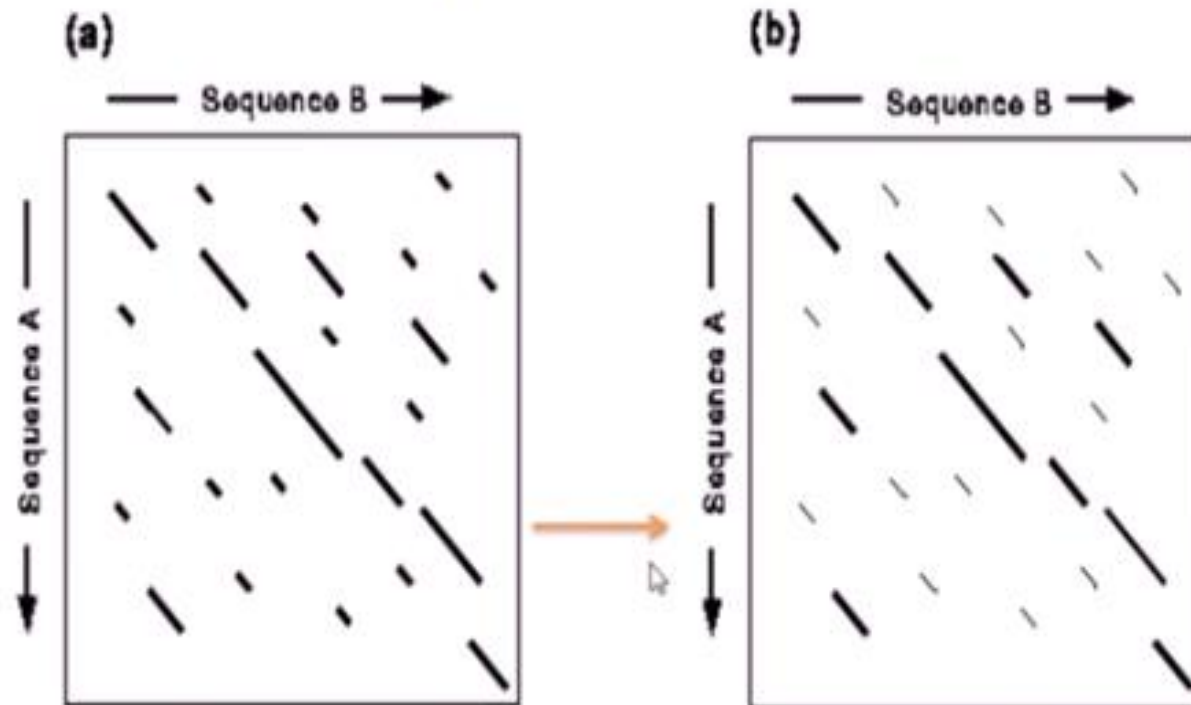
Introduction to bioinformatics, Autumn 2007

2. Score diagonals with k word match

We would like to find high scoring diagonals of dot matrix



3. Rescoring initial regions with substitution matrix (here PAM)



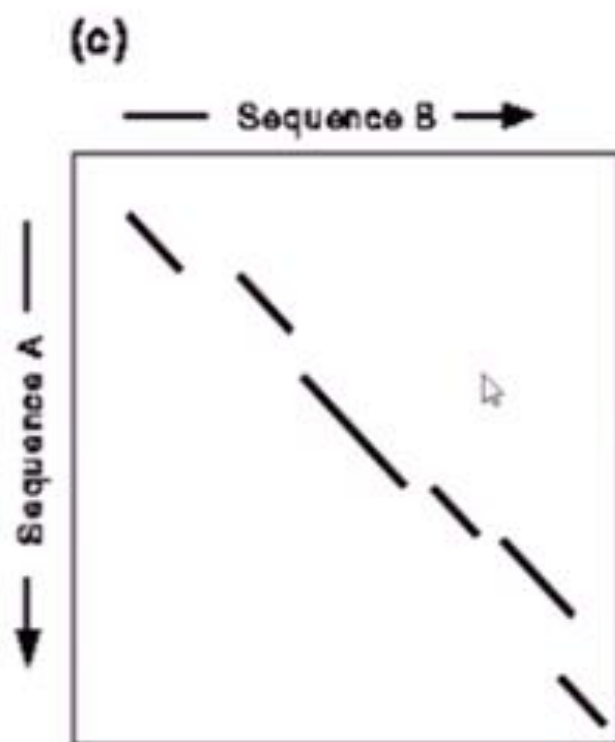
Find runs of identical words

Re-score using PAM matrix
Keep top scoring segments

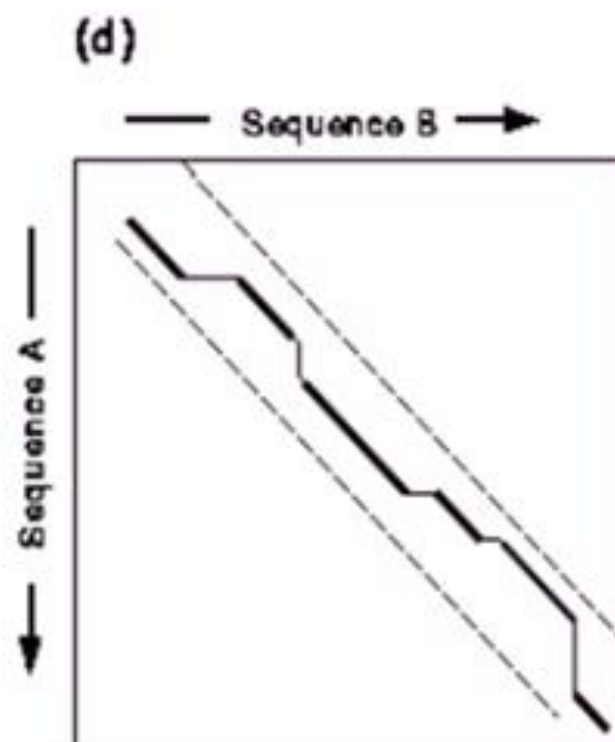
Initn: The best match after
rescoring

4. Join initial regions with gaps, penalize for gaps

After all diagonals are found, tries to join diagonals by adding gaps
Computes alignments in regions of best diagonals



Join segments using gaps,
eliminate other segments



Use dynamic programming to
create an optimal alignment

Dynamic Vs Heuristic

Aspect	Dynamic Programming	Heuristic Algorithms
Goal	Guarantees the optimal solution	Finds an approximate solution
Search Strategy	Exhaustive search over all possibilities	Focuses on high-probability regions
Computation Cost	Time and memory-intensive	Computationally efficient (faster)
Accuracy	Always accurate (optimal)	Accuracy may vary (suboptimal)
Use Cases	Small-scale problems or where precision is critical	Large-scale problems where speed is prioritized